Vineet Singh

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Overview

Software Engineer with 5 years of experience in developing and optimizing Linux-based system applications. Proficient in Python, C, and shell scripting, with a strong focus on user-space development, debugging, and performance tuning. Skilled in building robust, scalable solutions on Ubuntu platforms, and collaborating effectively in Agile environments.

Skills

- **Python:** System-level scripting, backend development (Flask, FastAPI), automation, testing (Pytest)
- **Linux:** OS internals, audio and Bluetooth stack management, udev rules, shell scripting, system configuration, daemon development
- **C Programming:** Low-level programming, kernel code debugging, legacy code refactoring
- **Operating Systems:** Ubuntu (22.04/24.04), FreeBSD, ThinOS customization and migration
- **Device Management:** USB device control via VID/PID and class codes, BLE/non-BLE Bluetooth automation
- **API Development and Testing:** RESTful APIs, Postman
- **DevOps Tools:** Jenkins, GitHub Actions, Docker, JFrog Artifactory, CI/CD pipelines
- **Version Control:** Git, GitHub, Bitbucket
- **Atlassian Tools:** Jira, Confluence
- **File Handling:** JSON, CSV, system logs
- **Other skills:** DSA, client facing

Experience

Software Engineer, Capgemini (Client: Dell Technologies) – Bengaluru, IN Nov 2022 – Till date

- Contributed to the migration of Dell ThinOS from FreeBSD to Ubuntu, ensuring system stability during the transition from Ubuntu 22.04 to 24.04 by resolving package compatibility issues and integrating updated dependencies.
- Developed and maintained system-level audio components including Jack, headset detection, Bluetooth profiles (A2DP/HSP), and auto-switching mechanisms using Python.
- Refactored legacy FreeBSD kernel-dependent code to align with Ubuntu architecture, improving maintainability and performance.
- Engineered device whitelisting/blacklisting logic using udev events, enabling dynamic USB device control based on VID/PID and class codes.
- Built a Bluetooth service to automate device pairing and reconnection, with support for BLE and non-BLE devices.
- Conducted in-depth debugging and performance profiling to enhance system daemon reliability and responsiveness.
- Collaborated in Agile development cycles, actively participating in sprint planning, code reviews, and retrospectives.

Python Developer, TSinfo Technology – BLR, IN Aug 2020 – Oct 2022

- Developed and optimized system utilities and automation scripts using Python for various enterprise use cases.
- Implemented logging, monitoring, and debugging tools to improve system reliability.
- Assisted in the implementation of OS-level features, enhancing system performance and usability.
- Developed a user interface using Tkinter and integrated it with a backend database, supporting both Oracle and SQLite.
- Technical documentation

Support Engineer, IQOR + Teleperformance, IN Feb 2018 – July 2019

Initially, worked as a support engineer for Symantec Antivirus, Hewlett Packard, and FedEx.

Projects

Gen AI Assistant Integration (July 2025)

- Explored multiple LLM APIs including OpenAI (ChatGPT) and Hugging Face, troubleshooting access and integration issues.
- Successfully set up and deployed Ollama (local LLM) on a Linux environment for offline inference.
- Developed Python scripts to interact with the Ollama API, tested endpoints using Postman, and integrated responses into a backend workflow.
- Gained hands-on experience with prompt engineering, API testing, and local model hosting.
- Used tools like Git for version control and Linux shell scripting for automation and environment setup. Environment: Python-OOPs, Flask framework, Linux operating system, Ollama model, Postman for fastAPI, Git for version controlling, Shell Scripting, PostgreSQL for database, restful API, FastAPI

DELL ThinOS Dell ThinOS

- Contributed in the migration from base os Ubuntu 22.04 to 24.04
- Addressed audio issues including Bluetooth, Jack and USB headsets
- Implemented blacklisting and whitelisting of USB devices to enhance security protocols. Technologies used: Udev, C, Python, File Systems, Alsa, PulseAudio

Dell Hybrid Client Dell Hybrid Client

- Integrated a robust audio server to improve system-wide audio handling and reliability.
- Audio Server Migration PipeWire -> PulseAudio
- Tools Used: Python, Linux, C Programming

Education

Chandigarh University, Master of Computer Application Aug 2019 – July 2021

- GPA: 7.5/10.0
- **Coursework:** Machine Learning, Data Structures and Algorithms, Computational Theory